Production deployment validation matrix

Production deployment validation matrix - 2.7.x upgrade

Status: **NOT READY FOR PRODUCTION** (see the gate rule).

Environment: development host `172.16.16.44` (`ocrv2`), release `2.7.0`, validated 12 November 2026.

Gate rule

Promote to production **only when every required row below is PASS** and the release owner has signed the bottom of this file. A single FAIL or NOT RUN blocks promotion. Optional rows do not block, but must be listed in the release notes.

Matrix

ID	Gate	Required	Status	Evidence
B1	Backend unit/integration tests	Yes	PASS	`pytest` : 290 passed, 179 skipped locally; CI runs the DB-backed set on PostgreSQL 17
B2	Frontend lint	Yes	PASS	0 errors, 132 warnings
B3	Frontend production build	Yes	PASS	`vite build` succeeds
B4	Lab validation	Yes	PASS	`validate-lab.py --all` : 169 labs, 0 errors, 0 warnings
B5	Supply-chain lock	Yes	PASS	42 files, 169 lab hashes match
B6	Licence compliance	Yes	PASS	`check-licences.py` : OK
B7	SPA security headers / CSP hash	Yes	PASS	`check-nginx-security-headers.py` , `check-csp-hash.py`
B8	External images in learner content	Yes	PASS	`check-external-images.py` : none
B9	Documentation links and assets	Yes	PASS	577 Markdown files, no broken links; assets PASS
B10	Compose config and image builds	Yes	PASS	Backend and frontend images build; compose config valid
B11	Alembic head	Yes	PASS	Dev DB at `0014_performance_indexes` ; startup upgrade clean
R1	Health and version on dev	Yes	PASS	`/health` 200, `/api/version` 2.7.0

ID	Gate	Required	Status	Evidence
R2	Connection-pool burst	Yes	PASS	36 concurrent dashboard/admin requests in 2s; 11/20 pool connections; no QueuePool errors
R3	Scheduler tick responsiveness	Yes	PASS	16/16 health samples at 200 through the 120s tick
R4	Software integrity endpoint	Yes	PASS	169/169 labs match; 0 mismatched/missing/extra; 0.2s
R5	CyberChef panel	Yes	PASS	`/tools/cyberchef/` 200, assets 200, inline-script hashes match CSP
R6	High-contrast theme	Yes	PASS	Theme cycle dark/light/contrast; contrast tokens present in the built bundle
R7	Mobile shell (student)	Yes	PASS	390x844: dashboard, exercises, profile, rangebox; 0 overflow, 0 console errors, tab bar present
R8	Mobile shell (admin/instructor)	Yes	PASS	390x844: dashboard, admin, instructor; notices present; 0 overflow/errors
R9	New OT lab (S7comm)	Yes	PASS	Seeded id 187, image prebuilt, DB1 read returns the flag
R10	Image upload hardening	Yes	PASS	7 tests: PNG accepted/re-encoded; SVG, HTML, mismatch, oversized rejected
D1	FK orphan scan (all FKs)	Yes	PASS	No orphan rows reported
D2	Nulls in critical columns	Yes	PASS	users/labs/levels/sessions/completions/settings: 0
D3	Duplicate usernames/emails/slugs	Yes	PASS	0
D4	Numeric sanity	Yes	PASS	No negative durations/counters, no future timestamps
D5	Ledger attribution	No	WARN	6,928 of 7,785 rows unattributed (seed/CLI/system writes); review queue noisy
D6	Stale lab rows	No	PASS	178 DB labs = 169 repo + 9 intentional `demotest-*`
P1	Backup restore drill	Yes	PASS	`pg_dump` + `pg_restore` into a scratch database; counts match exactly (141 users, 178 labs, 230 completions, 158 achievements)
P2	Upgrade rehearsal 2.6 -> 2.7	Yes	PASS	Scratch DB downgraded 0014 -> 0011 (2.6-era head) and upgraded back to 0014; row counts unchanged (141/178/230)
P3	Rollback rehearsal	Yes	PASS	Full `pg_dump` / `pg_restore` rollback plus the Alembic downgrade path above
P4	Signed release tag	Yes	NOT RUN	No release signing key is available in this environment; tag signing remains an operator step
P5	Production-shaped host rehearsal	Yes	NOT RUN	No second production-shaped host; dev uses the dev compose profile
P6	Email delivery	Yes	PASS	Mailpit sink on the compose network; `/api/settings/mail/test` delivered "SWOCTS Cyber Range mail test" to admin@example.com

ID	Gate	Required	Status	Evidence
P7	WireGuard / Peer Manager	Yes	PASS	Hardened build on `172.17.0.1:5000`; `/peers` returns 401 without a key and 200 with the backend key; `preshared_key` nulled
P8	RangeBox desktop launch	Yes	PASS	Spawned session 497 (`high-school-9-1-rangebox-first-steps`): running in 15s, RangeBox and portal containers up, stop initiated
P9	Exhaustive button/flow E2E	No	PARTIAL	Route-level Playwright smoke passed; not every button
P10	Dependency/image digest pinning	No	PARTIAL	Base images remain tag-pinned; digests reported at runtime

Result

- Required PASS: 31
- Required FAIL: 0
- Required NOT RUN: 2 (P4 signed tag, P5 production-shaped host)
- Warnings: 1 (D5)
- Completion: **31 of 33 required gates (94%)**

Decision: DO NOT PROMOTE. The application is healthy on dev, but the production-readiness gates for restore, upgrade/rollback rehearsal, signed release, and the infrastructure integrations are not complete.

Exit criteria for the next attempt

1. Run P1-P8 on a production-shaped host and record evidence in this file.
2. Cut and sign the release tag; update `docs/RELEASES.md`.
3. Either fix D5 (set actor context for seed/discovery/scheduler writes) or record an accepted-risk note with a retention policy for the ledger.
1. Re-run B1-B10 and R1-R10 on the release commit.
2. Only then change this file to `READY FOR PRODUCTION` and sign below.

Sign-off

Role	Name	Decision	Date
Release owner			
Data analyst		PASS on P1-P3; D5 conditional	2026-11-12
Security reviewer		PASS on P7; P4/P10 pending	2026-11-12